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Dynamic spectrum access mode based on station capabilities

Abstract

Dynamic spectrum access mode based on station capabilities is provided by categorizing functionalities of Access Points (APs) and stations (STA) in a wireless network; identifying interference induced by external signaling devices on channels in the wireless network; calculating an impact factor of the interference based on proximity of the external signaling devices to the wireless network, a pattern of the external signaling devices, and an extent of overlap with frequencies used by the external signaling devices and the wireless network; and in response to identifying a given STA that is paired with a given AP in the wireless network, wherein the given STA and the given AP are both categorized as being capable of both multilink communications and preamble puncturing, assigning network resources for the given STA to communicate with the given AP via one of multilink communications or preamble puncturing based on the impact factor of the interference.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. patent application Ser. No. 17/454,340 filed Nov. 10, 2021. The aforementioned related patent application is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Embodiments presented in this disclosure generally relate to wireless communications. More specifically, embodiments disclosed herein provide for improved bandwidth usage based on the capabilities of devices accessing a shared wireless network.

BACKGROUND

(2) In a wireless network, especially publically accessible networks, the various devices often have different capabilities, and may run hardware or software that are not using the same version one another. The wireless communications standards used by these devices, such as the IEEE 802.11 family of “Wi-Fi” standards, often offer backwards compatibility between different devices, so that devices running newer versions can communicate with devices running older versions of the standard. However, as communications standards progress, the devices using the newer standard often have additional capabilities that the devices using the older standard lack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate typical embodiments and are therefore not to be considered limiting; other equally effective embodiments are contemplated.
- (2) FIG. 1 illustrates a network environment, according to embodiments of the present disclosure.
- (3) FIG. 2 illustrates differently multilink operations possible in the wireless network, according to embodiments of the present disclosure.
- (4) FIG. 3 illustrates channel assignment, according to embodiments of the present disclosure.
- (5) FIG. 4 is a flowchart for a method to provide dynamic spectrum access mode based on station capabilities, according to embodiments of the present disclosure.
- (6) FIG. 5 illustrates hardware of a computing device, according to embodiments of the present disclosure.
- (7) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially used in other embodiments without specific recitation.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

- (8) One embodiment presented in this disclosure is a method, comprising: categorizing functionalities of Access Points (APs) and mobile stations (STA) in a wireless network; identifying interference induced by external signaling devices on channels used by the wireless network; calculating an impact factor of the interference based on a proximity of the external signaling devices to the wireless network, a pattern of the external signaling devices, and an extent of overlap with frequencies used by the external signaling devices and the wireless network; and in response to identifying a given STA that is paired with a given AP in the wireless network, wherein the given STA and the given AP are both categorized as being capable of both multilink communications and preamble puncturing, assigning network resources for the given STA to communicate with the given AP via one of multilink communications or preamble puncturing based on the impact factor of the interference.
- (9) One embodiment presented in this disclosure is a system, comprising: a processor; and a memory including instructions that when performed by the processor configure the process to perform operations comprising: categorizing functionalities of Access Points (APs) and mobile stations (STAs) in a wireless network; identifying interference induced by external signaling devices on channels used by the wireless network; calculating an impact factor of the interference based on a proximity of the external signaling devices to the wireless network, a duty cycle of the external signaling devices, and an extent of overlap with frequencies used by the external signaling devices and the wireless network; and in response to identifying a given STA that is paired with a

given AP in the wireless network, wherein the given STA and the given AP are both categorized as being capable of both multilink communications and preamble puncturing, assigning network resources for the given STA to communicate with the given AP via one of multilink communications or preamble puncturing based on the impact factor of the interference.

(10) One embodiment presented in this disclosure is a non-transitory computer readable storage medium storing instructions that when executed by a processor perform operations comprising: categorizing functionalities of Access Points (APs) and mobile stations (STAs) in a wireless network; identifying interference induced by external signaling devices on channels used by the wireless network; calculating an impact factor of the interference based on a proximity of the external signaling devices to the wireless network, a duty cycle of the external signaling devices, and an extent of overlap with frequencies used by the external signaling devices and the wireless network; and in response to identifying a given STA that is paired with a given AP in the wireless network, wherein the given STA and the given AP are both categorized as being capable of both multilink communications and preamble puncturing, assigning network resources for the given STA to communicate with the given AP via one of multilink communications or preamble puncturing based on the impact factor of the interference.

Example Embodiments

(11) The present disclosure relates to dynamic spectrum access to allow greater use of bandwidth in a network environment. User Equipment (UE) and Access Points (APs) in the environment are evaluated to determine the associated signaling capabilities for the devices and whether different access modes would provide better or more efficient use of the available bandwidth. Depending on the capabilities of the UEs and APs in the environment (e.g., what the UEs and APs are capable or incapable of performing), and the competing sources for the available bandwidth, a multilink operation or a preamble puncturing operation can be biased for selection when scheduling communications. As users may delay deployment of devices with the latest capabilities, categorizing the functionalities of the actual devices connected to or providing the network, and biasing operation of the network in accordance with those functionalities thereby allow a network provider to more efficiently allocate bandwidth and channel assignment in preparation for known interference sources and competing signaling devices.

(12) FIG. 1 illustrates a network environment **100**, according to embodiments of the present disclosure. In FIG. 1, one or more Access Points (APs) **110a-d** (generally or collectively, AP **110**) provide a wireless network **120** to various mobile stations (STA) **130a-b** (generally or collectively STA **130**) in the environment **100**. In various aspects, the wireless network **120** is a cellular or Wi-Fi based network offered to users in a public or private venue. In various embodiments, the wireless network **120** includes a network controller **140** that communicates with the APs **110** to coordinate network management among the APs **110**, although the APs **110** can also manage the network among themselves, thus omitting the network controller **140** in some embodiments.

(13) The APs **110** can offer wireless communication sessions in the wireless network **120** according to various Radio Access Technologies (RAT) and communications standards such as, but not limited to, “Wi-Fi” networking according to the various families, sub-standards, and derivatives of the IEEE 802.11 standard, cellular networking including various generations and subtypes thereof, such as, Long Term Evolution (LTE) and Fifth Generation New Radio (5G NR) networks, Citizens Broadband Radio Service (CBRS) networks, or the like. Example hardware as may be included in an AP **110** is discussed in greater detail in regard to FIG. 5.

(14) The STAs **130** may include any computing device that can wirelessly connect to one or more APs **110**. Example STAs **130** can include, but are not limited to: smart phones, feature phones, tablet computers, laptop computers, desktop computers, Internet of Things (IoT) devices, and the like. In various embodiments, the STAs **130** can also be referred to as a User Equipment (UE), a client device (CD), a user device, or an endpoint. Example hardware as may be included in a STA **130** is discussed in greater detail in regard to FIG. 5.

(15) The network controller **140**, if included, may include any computing device or cloud based service that is configured to interface with two or more APs **110** to coordinate how spectrum and services are shared in the environment **100**. The network controller **140** can be provided on a separate computing device connected to the individual APs **110** via wired or wireless communications, may be included with a “central” or “commander” AP **110**, or may be provided in an ad hoc arrangement via a collective of two or more APs **110** negotiating among themselves for network management. Accordingly, any action ascribed to the network controller **140** in an example given in the present disclosure may also or instead be ascribed to one or more of the APs **110**. Example hardware as may be included in a network controller **140** is discussed in greater detail in regard to FIG. 5.

(16) In addition to the members of the wireless network **120**, various external signaling devices **150a-c** (generally or collectively, external signaling device **150**) may consume or compete for the bandwidth used in the wireless network **120**. For example, devices in neighboring wireless networks under the control of a different party, may seek to use some or all of the bandwidth used by the wireless network **120** according to the same family of communications standards (e.g., as part of an Overlapping Basic Service Set (OBSS)). These devices using the same communications standard can include managed neighbors **150a** in a shared or partnered network that the network controller **140** can negotiate with for access to the shared spectrum or for pre-planning when the other network will use the shared spectrum, or rogue APs **150b** that do not coordinate with the network controller **140** for access to the shared spectrum (e.g., leaving the network controller **140** to react to the spectrum use choices of the rogue AP **150b**).

(17) In external signaling devices that use the same family of communications standards as the wireless network **120**, various incumbent signaling devices **150c** with higher priority or legacy access rights to the spectrum can use the shared spectrum according to a different signaling standard (for communications or other purposes). For example, a weather station may use some or all of the bandwidth used by the wireless network **120** for Doppler signals (e.g., a different transmission standard than the wireless network **120**). In various embodiments, the network controller **140** may defer to incumbent users with higher-priority access to the share bandwidth. When deferring access to an incumbent signaling device **150c** (e.g., such as a weather station), where the network controller **140** abandons or yields those portions of the spectrum to the external signaling devices **150** when a legacy communications standard has higher-priority access to the shared portions of spectrum, allows the incumbent signaling device **150c** full access to the shared channel for a least a given time period.

(18) FIG. 2 illustrates differently multilink operations possible in the wireless network **120**, according to embodiments of the present disclosure.

(19) A first STA **130a** is shown in simultaneous transmit and receive (STR) multilink communications with a first AP **110a**. In an STR multilink communications arrangement, a STA **130** is capable of sending uplink traffic to an associated AP **110** on a first channel and receiving downlink traffic from the AP **110** on a second channel at the same time. Although shown using two channels, in various embodiments, a STR-capable pair of STAs **130** and APs **110** may use more than two channels for STR multilink communications, with various numbers of uplink or downlink channels.

(20) A second STA **130b** and a third STA **130c** are shown in non-STR multilink communications with the first AP **110a**. In non-STR multilink communications arrangements, a STA **130** uses two or more channels to communicate with an associated AP **110**, but engages in unidirectional communications. For example, the second STA **130b** uses two or more channels for uplink communications at the same time (but no channels for downlink communications) while the third STA **130c** uses two or more channels for downlink communications at the same time (but no channels for uplink communications). In various embodiments, a STA **130** engaged in non-STR multilink communications may be capable of STR multilink communications, but is scheduled by

the associated AP **110** (e.g., due to traffic queueing) for non-STR communications in a given timeslot; however, some STA **130** engaged in non-STR multilink communications may be capable of only unidirectional multilink communications. Although shown using two channels, in various embodiments, a non-STR-capable pair of STAs **130** and APs **110** may use more than two channels for non-STR multilink communications, with various numbers of uplink or downlink channels.

(21) A fourth STA **130d** is shown engaged in Enhanced Single Radio (ESR) communications, where the fourth STA **130** dynamically switches a single radio between communicating with one of the first AP **110a** and a second AP **110b** at a given time. The fourth STA **130d** is within range of two or more APs **110**, and in various embodiments can send uplink communications to one or more of the APs **110** and receive downlink communication from one or more of the AP **110**. Each AP **110** associates one channel with the STA **130**. Although shown using two channels with two associated APs **110**, in various embodiments, a ESR-capable STAs **130** may be in communication with two or more APs **110** (and associated channels) for ESR communications, with various numbers of uplink or downlink channels.

(22) FIG. 3 illustrates channel assignment, including preamble puncturing, multilink communication, or full channel assignment, according to embodiments of the present disclosure. A macro-channel **310** spans a frequency band from $\lambda.\text{sub}.0$ to $\lambda.\text{sub}.4$, while four micro-channels **320a-d** (generally or collectively, micro-channel **320**) span frequency bands from $\lambda.\text{sub}.0$ to $\lambda.\text{sub}.1$, $\lambda.\text{sub}.1$ to $\lambda.\text{sub}.2$, $\lambda.\text{sub}.2$ to $\lambda.\text{sub}.3$, and $\lambda.\text{sub}.3$ to $\lambda.\text{sub}.4$. The micro-channels **320** when combined occupy the same bandwidth as a macro-channel **310**. For example, a macro-channel **310** of 80 Megahertz (MHz) has the same bandwidth as four micro-channels **320a-d** of 20 MHz. When overlapped, the micro-channels **320** occupy the same frequency band as the macro-channel **310**, but may describe different named channels. For example, the four micro-channels **320a-d** may be defined on channels **52**, **56**, **60**, and **64**, while the macro-channel **310** is defined on channel **58** (to encompass all of channels **52**, **56**, **60**, and **64**).

(23) As used herein, the lowest frequency micro-channel **320** of a set of micro-channels **320** may be referred to as a primary channel, and the other micro-channels **320** may be referred to as secondary channels. In the current examples, the first micro-channel **320a** would be the primary channel, and the second through fourth sub-channels **320b-d** would be secondary channels.

(24) When used individually, each micro-channel **320** may be assigned to different STAs **130** or to one STA **130** for multilink operations. Each transmission over a micro-channel **320** includes various overhead sections (e.g., header fields, checksum fields) and payload data, as does a transmission over a macro-channel **310**. Accordingly, as the macro-channel **310** includes one set of overhead section for the same amount of bandwidth as the collective micro-channels **320** in the same frequency band, the macro-channel **310** can offer greater efficiency for data throughput. However, the micro-channels **320** can offer greater flexibility in channel assignment and (potentially) for interference mitigation or avoidance compared to the associated macro-channel **310**.

(25) In various embodiments, the AP **110** can change the width of a channel assigned for communications for a STA **130** on a frame-by-frame basis, so that a STA **130** may initially use the macro-channel **310**, and switch to using one or more of the micro-channels **320** if interference **330** is detected between $\lambda.\text{sub}.0$ and $\lambda.\text{sub}.4$. Depending on the frequencies that the interference **330** spans, how long or how often the interference **330** is observed, and the capabilities of the AP **110** and the STA **130**, when a channel assignment is changed from the macro-channel **310** to the micro-channels **320**, various different spectrum assignments may be considered.

(26) When the AP **110** and STA **130** support multilink communications, the AP **110** can assign multiple micro-channels **320** for communication with the STA **130**. For example, when the interference **330** is located in the second micro-channel **320b**, the AP **110** may assign one or more of the remaining micro-channels **320** to the STA **130**. For example, when the AP **110** and the STA **130** do not support multilink communications, the AP **110** may assign one of a first free frame **340a**

(generally or collectively, free frames **340**) in the first micro-channel **310a**, a second free frame **340b** in the third micro-channel **320c**, or a third free frame **340c** in the fourth micro-channel for use to the STA **130**. When the AP **110** and the STA **130** do support multilink communications, the AP **110** may assign one or more of the free frames **340** to the STA **130**. In various embodiments, the free frames **340** may be assigned for uplink communications or downlink communications, and (if supported) when two or more free frames are assigned, some may be used for uplink while some are used simultaneously for downlink communications.

(27) In various embodiments, depending on the frequency filtering capabilities of the STA **130**, some of the free frames **340** may be unavailable, despite being free of interference. Accordingly, the capabilities of the STA **130** can effectively allow the interference **330** to bleed over into adjacent frequency bands when guard frequencies, filters, or channel spacing requirements prevent the use of channels adjacent to one another or adjacent to interference **330**. For example, a STA **130** using a filter that cannot differentiate the interference **330** in the second micro-channel **320b** from the adjacent micro-channels **320**, those adjacent micro-channels **320** may be unavailable for use with free frames **340**, despite the interference **330** not overlapping the associated micro-channels **320**. However, the affected STA **130** may still be able to use the fourth micro-channel **320d** for communications, as the filter of the STA **130** has sufficient discretion to prevent bleed over from the second micro-channel **320b** to the fourth micro-channel **320d**.

(28) When the AP **110** and STA **130** support preamble puncturing, instead of falling back to using the micro-channels **320**, the AP **110** can allow the STA **130** to transmit on the primary channel and any of the secondary channels not overlapped with the interference **330** as one coherent punctured frame **350**. For example, when the interference **330** overlaps with the frequencies used by the second micro-channel **320b**, the AP **110** can assign a punctured frame **350** to the STA **130** that omits frequencies $\lambda_{\text{sub.1}}$ to $\lambda_{\text{sub.2}}$ but includes frequencies from $\lambda_{\text{sub.0}}$ to $\lambda_{\text{sub.1}}$ and $\lambda_{\text{sub.2}}$ to $\lambda_{\text{sub.4}}$. Although discussed in relation to interference that overlaps the second micro-channel **320b**, preamble puncturing can be used when a different secondary channel or more than one secondary channel **302** is overlapped.

(29) Once the interference is no longer present, the AP **110** may return to assigning a full frame **360** that covers the macro-channel **310**.

(30) Accordingly, the network controller **140** monitors the external signaling devices **150** to identify when interference **330** is generated and which channels that interference **330** affects as part of determining whether to use punctured frames **350** for the STAs **130** capable of benefiting from preamble puncturing.

(31) FIG. **4** is a flowchart for a method **400** to provide dynamic spectrum access mode based on station capabilities, according to embodiments of the present disclosure. Method **400** begins a block **410**, where the network controller **140** categorizes the functionalities of the member devices in the wireless network **120**. In various embodiments, the APs **110** that are members of the wireless network **120** are categorized based on whether the AP **110** supports multilink operations (including STR, non-STR, or ESR-supported multilink communications), preamble puncturing, both multilink operations and preamble puncturing, or neither multilink operations nor preamble puncturing. Similarly, the network controller **140** categorizes the STAs **130** that are connected to the APs **110**, where the STAs **130** are categorized based on whether the STAs **130** are capable of STR multilink communications, non-STR multilink communications, ESR communications, or preamble puncturing, or combinations thereof.

(32) At block **420**, the network controller **140** identifies external interference on the wireless network **120**, such as may be induced by various external signaling devices **150** on one or more channels used by the wireless network **120**. In various embodiments, the network controller **140** identifies the interference sources as radios from OBSSs (either as managed neighbors **150a** or rogue APs **150b**) using the same or compatible communications standards as the wireless network **120**, or as high-priority incumbent signaling devices **150c** (e.g., Doppler weather stations) that use

different signaling standards than the AP **110**. The network controller **140** identifies the characteristics of the interference generated by the various sources, indicating the channels affected, duration and signal strength of the interference, how often the interference is observed (e.g., the periodicity/frequency of the interference), and the like.

(33) At block **430**, the network controller **140** calculates an impact factor of the interference on the operations of STAs **130** within the wireless network **120**. The effect of the interference may vary based on what channels are affected, how often those channels are affected, and how the various STAs **130** can work around the interference. For example, interference on a primary channel may prevent STAs **130** from using preamble puncturing, while interference on a secondary channel may prevent STAs **130** from using that secondary channel, but still permits the use of preamble puncturing (so long as the affected channel is avoided). In another example, a first STA **130a** may be specified (e.g., by a manufacturer) to use signal separation of at least 20 MHz, so that 20 MHz channels adjacent to an interference-prone 20 MHz channel are not usable, despite being free of interference, while a second STA **130b** specified to use a lower signal separation may be able to use those adjacent channels. Accordingly, the impact factor identifies portions of the bandwidth that are usable by the individual STAs **130**.

(34) In addition to identifying the bandwidth assignment schemes for the individual STAs **130**, the impact factor evaluates the overall effect of one or more interference sources on the collective (one or more) STAs **130** in the wireless network **120**. Accordingly, the overall effect of the interference across the wireless network **120** can result in one of preamble puncturing or multilink operations, if applied solely to the signaling operations for the STAs **130**, being better for overall network stability, while individually worse for a non-majority share of the STAs **130**. Therefore, the impact factor identifies the numbers of STAs **130** that support different formats of communications (e.g., STR or non-STR multilink communication vs. ESR or no multilink communications, multilink communications vs. preamble puncturing) so that the network controller **140** assign channels based on which set of STAs **130** has more members. In various embodiments, a given STA **130** may belong to two or more groups (e.g., capable of both multilink operations and preamble puncturing), and would therefore be counted as a member of both groups for purposes of determining the relative sizes of different potential groups of STAs **130**.

(35) At block **440**, the network controller **140** biases operations of the wireless network **120** to provide high-bandwidth communications in light of the interference and the calculated impact factor. The biased operations seek to assign the available bandwidth for the greatest amount of use by the STAs **130** when operating in a potentially interference-prone environment. To that end, operations during interference-free time periods are biased for easy and rapid transition to operation during time periods that include interference.

(36) For example, if interference is known to affect a primary channel, the network controller **140** may bias channel assignment during interference-free time periods away from using a full frame **360** over the macro-channel **310**, as preamble puncturing is not available when interference **330** is present. Additionally or alternatively, when a STA **130** is not capable of preamble puncturing (either due to lack of functionality or channel separation specifications for the STA **130**), the network controller **140** may bias channel assignment to assign the individual STA **130** to a set of channels that are unaffected by interference or that the individual STA **130** can work around to continue transmitting in a subset of the unaffected micro-channels **320** with free frames **340** when interference **330** is present.

(37) In another example, if interference is known to affect secondary channels (and channel separation specifications for the STAs **130** allow the use of the adjacent channels), the network controller **140** may bias channel assignment during interference free-time periods towards using full frames **360** in the macro-channels **310**, as using a single macro-channel **310** is often more bandwidth efficient than using an equivalent bandwidth of micro-channels **320**. Accordingly, when interference **330** is detected, the assigned STA **130** may switch from using a full frame **360** in the

macro-channel **310** to using a punctured frame **350** that avoids the micro-channel(s) **320** affected by the interference **330**.

(38) Accordingly, by biasing the assignment of network channels to the various STA **130** based on the functionalities of those STA **130**, and the characteristics of the interference **330** in the wireless network **120**, the network controller **140** can switch the STAs **130** between interference-free and interference-mitigation modes of operation with lower change over effects, thereby improving the efficiency of the wireless network **120** (e.g., requiring fewer channel reassignments) and improving of use of available bandwidth in the environment **100**.

(39) At block **450**, the network controller **140** assigns network resources for the radios in the APs **110** and the STAs **130** based on the bias for multilink operation or preamble puncturing when interference is present. Accordingly, the network controller **140** assigns network resources for each STA **130** to communicate with the associated APs via one of multilink communications or preamble puncturing based on the impact factor of the interference **330**. The network resources may include various time frames that indicate which channels the STA **130** are to use and what mode to operate in when interference is detected (or expected) and which channels the STA **130** are to use and what mode to operate in when interference is not detected (or expected).

(40) For example, a first STA **130a** that is capable of operating in preamble puncturing mode may be assigned a macro-channel **310** to use with full frames **360** for a first time period, and transitions to use a punctured frame **350** when interference **330** is expected or detected. In contrast, a second STA **130b** that is capable of operating in multilink mode may be assigned (at a different time) the same channel as the first STA **130a** was assigned to, and uses several micro-channels **320** to send or receive data using several free frames **340** (whether in STR or non-STR mode) and stops using one or more of the micro-channels **320** when interference **330** is present on those channels, but continues using the non-overlapped micro-channels **320**.

(41) The network controller **140** applies the bias for assignment of the available channels mitigates the effects of the interference while providing high-bandwidth communications according to the impact factor. Method **400** may conclude at block **450**, and may repeat at a predefined time interval, in response to changes in the interference sources, in response to STAs **130** joining or leaving the wireless network **120**, or changes in the deployment of the APs **110**.

(42) FIG. 5 illustrates hardware of a computing device **500** such as can be included in an AP **110**, STA **130**, or network controller **140** as described herein. The computing device **500** includes a processor **510**, a memory **520**, and communication interfaces **530**. The processor **510** may be any processing element capable of performing the functions described herein. The processor **510** represents a single processor, multiple processors, a processor with multiple cores, and combinations thereof. The communication interfaces **530** facilitate communications between the computing device **500** and other devices. The communication interfaces **530** are representative of wireless communications antennas (both omnidirectional and directional), various steering mechanisms for the antennas, and various wired communication ports including out-pins and in-pins to a microcontroller. The memory **520** may be either volatile or non-volatile memory and may include RAM, flash, cache, disk drives, and other computer readable memory storage devices. Although shown as a single entity, the memory **520** may be divided into different memory storage elements such as RAM and one or more hard disk drives.

(43) As shown, the memory **520** includes various instructions that are executable by the processor **510** to provide an operating system **521** to manage various functions of the computing device **500** and one or more applications **522** to provide various functionalities to users of the computing device **500**, which include one or more of the functions and functionalities described in the present disclosure.

(44) In the current disclosure, reference is made to various embodiments. However, the scope of the present disclosure is not limited to specific described embodiments. Instead, any combination of the described features and elements, whether related to different embodiments or not, is

contemplated to implement and practice contemplated embodiments. Additionally, when elements of the embodiments are described in the form of “at least one of A and B,” it will be understood that embodiments including element A exclusively, including element B exclusively, and including element A and B are each contemplated. Furthermore, although some embodiments disclosed herein may achieve advantages over other possible solutions or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the scope of the present disclosure. Thus, the aspects, features, embodiments and advantages disclosed herein are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the invention” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

(45) As will be appreciated by one skilled in the art, the embodiments disclosed herein may be embodied as a system, method or computer program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.”

Furthermore, embodiments may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(46) Program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(47) Computer program code for carrying out operations for embodiments of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(48) Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems), and computer program products according to embodiments presented in this disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

(49) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other device to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the block(s) of the flowchart illustrations and/or block diagrams.

(50) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process such that the instructions which execute on the computer, other programmable data

processing apparatus, or other device provide processes for implementing the functions/acts specified in the block(s) of the flowchart illustrations and/or block diagrams.

(51) The flowchart illustrations and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments. In this regard, each block in the flowchart illustrations or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(52) In view of the foregoing, the scope of the present disclosure is determined by the claims that follow.

Claims

1. A method, comprising: identifying interference induced by external signaling devices on channels used by a wireless network; calculating an impact factor of the interference on at least one of performing preamble puncturing or multilink operation in the wireless network; and assigning network resources for a given station (STA) to communicate with a given access point (AP) based on the impact factor of the interference and functionalities of the given STA and the given AP.
2. The method of claim 1, further comprising: biasing assignment of the channels in the wireless network to provide a first set of STAs that are incapable of multilink operation or preamble puncturing to a first set of channels unaffected by the interference and to provide a second set of STAs that are capable of one or more of multilink operation and preamble puncturing to a second set of channels at least partially affected by the interference based on the functionalities of the STAs.
3. The method of claim 1, further comprising: biasing assignment of the channels in the wireless network to provide multilink operations for a first set of STAs capable of Enhanced Single Radio (ESR) multilink communications over preamble puncturing and to provide a second set of STAs that support multilink communications with preamble puncturing over multilink communications.
4. The method of claim 1, further comprising: identifying a first number of STAs that support STR or non-STR multilink communications and a second number of STAs that support ESR or do not support multilink communications; and biasing assignment of the channels for preamble puncturing when the first number is higher than the second number or for multilink communications when the second number is higher than the first number.
5. The method of claim 1, wherein the impact factor identifies a portion of usable bandwidth in a given channel over time when the given channel is affected by the interference, wherein the portion of the usable bandwidth is identified as including or excluding a primary channel for multilink operation.
6. The method of claim 1, further comprising: categorizing a plurality of APs as one of: supporting both multilink operations and preamble puncturing; supporting one of multilink operations and preamble puncturing; and supporting neither multilink operations nor preamble puncturing; and categorizing a plurality of STAs as at least one of: capable of simultaneous transmit and receive (STR) multilink communications; capable of non-STR multilink communications; capable of Enhanced Single Radio (ESR) communications to dynamically switch a single radio between

communicating with two or more APs; capable of preamble puncturing; and supporting neither multilink operations nor preamble puncturing.

7. The method of claim 1, wherein the external signaling devices include at least one of: radios from an Overlapping Basic Service Set (OBSS) that are classified into one of: managed neighbors in a shared network with the wireless network; or rogue APs that are not included in the shared network; or incumbent high-priority devices using a different signaling standard than the given AP.

8. A system, comprising: a processor; and a memory including instructions that when performed by the processor configure the processor to perform operations comprising: identifying interference induced by external signaling devices on channels used by a wireless network; calculating an impact factor of the interference on at least one of performing preamble puncturing or multilink operation in the wireless network; and assigning network resources for a given station (STA) to communicate with a given access point (AP) based on the impact factor of the interference and functionalities of the given STA and the given AP.

9. The system of claim 8, wherein the operations further comprise: biasing assignment of the channels in the wireless network to provide a first set of STAs that are incapable of multilink operation or preamble puncturing to a first set of channels unaffected by the interference and to provide a second set of STAs that are capable of one or more of multilink operation and preamble puncturing to a second set of channels at least partially affected by the interference based on the functionalities of the STAs.

10. The system of claim 8, wherein the operations further comprise: biasing assignment of the channels in the wireless network to provide multilink operations for a first set of STAs capable of Enhanced Single Radio (ESR) multilink communications over preamble puncturing and to provide a second set of STAs that support multilink communications with preamble puncturing over multilink communications.

11. The system of claim 8, wherein the operations further comprise: identifying a first number of STAs that support STR or non-STR multilink communications and a second number of STAs that support ESR or do not support multilink communications; and biasing assignment of the channels for preamble puncturing when the first number is higher than the second number or for multilink communications when the second number is higher than the first number.

12. The system of claim 8, wherein the impact factor identifies a portion of usable bandwidth in a given channel over time when the given channel is affected by the interference, wherein the portion of the usable bandwidth is identified as including or excluding a primary channel for multilink operation.

13. The system of claim 8, wherein: the given AP is categorized as one of: supporting both multilink operations and preamble puncturing; supporting one of multilink operations and preamble puncturing; and supporting neither multilink operations nor preamble puncturing; and the given STA is categorized as at least one of: capable of simultaneous transmit and receive (STR) multilink communications; capable of non-STR multilink communications; capable of Enhanced Single Radio (ESR) communications to dynamically switch a single radio between communicating with two or more APs; capable of preamble puncturing; and supporting neither multilink operations nor preamble puncturing.

14. The system of claim 8, wherein the external signaling devices include at least one of: radios from an Overlapping Basic Service Set (OBSS) that are classified into one of: managed neighbors in a shared network with the wireless network; or rogue APs that are not included in the shared network; or incumbent high-priority devices using a different signaling standard than the given AP.

15. A non-transitory computer readable storage medium storing instructions that when executed by a processor perform operations comprising: identifying interference induced by external signaling devices on channels used by a wireless network; calculating an impact factor of the interference on at least one of performing preamble puncturing or multilink operation in the wireless network; and assigning network resources for a given station (STA) to communicate with a given access point

(AP) based on the impact factor of the interference and functionalities of the given STA and the given AP.

16. The non-transitory computer readable storage medium of claim 15, wherein the operations further comprise: biasing assignment of the channels in the wireless network to provide a first set of STAs that are incapable of multilink operation or preamble puncturing to a first set of channels unaffected by the interference and to provide a second set of STAs that are capable of one or more of multilink operation and preamble puncturing to a second set of channels at least partially affected by the interference based on the functionalities of the STAs.

17. The non-transitory computer readable storage medium of claim 15, wherein the operations further comprise: biasing assignment of the channels in the wireless network to provide multilink operations for a first set of STAs capable of Enhanced Single Radio (ESR) multilink communications over preamble puncturing and to provide a second set of STAs that support multilink communications with preamble puncturing over multilink communications.

18. The non-transitory computer readable storage medium of claim 15, wherein the operations further comprise: identifying a first number of STAs that support STR or non-STR multilink communications and a second number of STAs that support ESR or do not support multilink communications; and biasing assignment of the channels for preamble puncturing when the first number is higher than the second number or for multilink communications when the second number is higher than the first number.

19. The non-transitory computer readable storage medium of claim 15, wherein the impact factor identifies a portion of usable bandwidth in a given channel over time when the given channel is affected by the interference, wherein the portion of the usable bandwidth is identified as including or excluding a primary channel for multilink operation.

20. The non-transitory computer readable storage medium of claim 15, wherein the operations further comprise: categorizing a plurality of APs as one of: supporting both multilink operations and preamble puncturing; supporting one of multilink operations and preamble puncturing; and supporting neither multilink operations nor preamble puncturing; and categorizing a plurality of STAs at least one of: capable of simultaneous transmit and receive (STR) multilink communications; capable of non-STR multilink communications; capable of Enhanced Single Radio (ESR) communications to dynamically switch a single radio between communicating with two or more APs; capable of preamble puncturing; and supporting neither multilink operations nor preamble puncturing.
